Onboarding data from Linux Server

Summary

This document describes the steps involved in Installing Universal Forwarder on Linux Server Endpoint and configuring forwarding and monitoring of those logs from the Search Head UI Component of Splunk.

1. Install Universal Forwarder in Linux Server

Refer **07a_Install_splunk_UF_Linux_server_V1.pdf**.

2. Connect Universal Forwarder with Intermediate Heavy Forwarder

Refer **07b_Connect_UF_with_HF_V1.pdf**.

3. Switch to the splunk user and verify if the files are readable

```
sudo su - splunk
cat /var/log/messages
cat /var/log/audit/audit.log
exit
```

```
[ec2-user@LinuxServer ~]$ sudo su - splunk
Last login: Tue Apr  7 01:35:07 UTC 2026 on pts/0
[splunk@LinuxServer ~]$ cat /var/log/messages
cat: /var/log/messages: Permission denied
[splunk@LinuxServer ~]$ cat /var/log/audit/audit.log
cat: /var/log/audit/audit.log: Permission denied
[splunk@LinuxServer ~]$
```

Note: If you get a **"Permission denied"** error, proceed to the next step to grant read permissions.

4. Switch to the root user and run the following commands:

```
sudo su
chmod o+r /var/log/messages
chmod o+r /var/log/audit/audit.log
```

```
[ec2-user@LinuxServer ~]$ sudo su
[root@LinuxServer ec2-user]# chmod o+r /var/log/messages
[root@LinuxServer ec2-user]# chmod o+r /var/log/audit/audit.log
[root@LinuxServer ec2-user]# ll /var/log/
total 784
drwx------. 2 root   root       23 Mar 18 03:45 audit
-rw-rw----. 1 root   utmp     1536 Apr  7 01:27 btmp
-rw-r--r--. 1 root   root      678 Apr  7 01:47 choose_repo.log
drwxr-x---. 2 chrony chrony      6 Nov 11  2024 chrony
-rw-r-----. 1 root   root   369145 Apr  7 00:43 cloud-init.log
-rw-r-----. 1 root   adm      9015 Apr  7 00:43 cloud-init-output.log
-rw-------. 1 root   root     2758 Apr  7 02:07 cron
-rw-r--r--. 1 root   root     6302 Apr  7 01:47 dnf.librepo.log
-rw-r--r--. 1 root   root    11312 Apr  7 01:47 dnf.log
-rw-r--r--. 1 root   root      787 Apr  7 01:47 dnf.rpm.log
-rw-r--r--. 1 root   root      540 Apr  7 01:47 hawkey.log
drwx------. 2 root   root        6 Aug 19  2024 insights-client
-rw-rw-r--. 1 root   utmp   292584 Apr  7 02:11 lastlog
-rw-------. 1 root   root        0 Dec  9 15:46 maillog
-rw----r--. 1 root   root   342255 Apr  7 02:11 messages
drwx------. 2 root   root        6 Dec  9 15:46 private
lrwxrwxrwx. 1 root   root       39 Dec  9 15:46 README -> ../../usr/share/doc/systemd/README.logs
drwxr-xr-x. 2 root   root       43 Mar 18 03:45 rhsm
-rw-------. 1 root   root    11514 Apr  7 02:11 secure
-rw-------. 1 root   root        0 Dec  9 15:46 spooler
drwxr-x---. 2 sssd   sssd        6 Oct 17 18:34 sssd
-rw-------. 1 root   root        0 Dec  9 15:46 tallylog
drwxr-xr-x. 2 root   root       23 Mar 18 03:45 tuned
-rw-rw-r--. 1 root   utmp     9600 Apr  7 00:45 wtmp
[root@LinuxServer ec2-user]# ll /var/log/audit/
total 184
-rw----r--. 1 root root 185972 Apr  7 02:13 audit.log
[root@LinuxServer ec2-user]#
```

5. Switch back to the Splunk user and confirm that the files are now readable

```
sudo su – splunk
cat /var/log/messages
```

```
Apr  7 01:54:33 LinuxServer systemd[1]: nm-cloud-setup.service: Deactivated successfully.
Apr  7 01:54:33 LinuxServer systemd[1]: Finished Automatically configure NetworkManager in cloud.
Apr  7 02:00:33 LinuxServer systemd[1]: Starting Automatically configure NetworkManager in cloud..
.
Apr  7 02:00:33 LinuxServer systemd[1]: nm-cloud-setup.service: Deactivated successfully.
Apr  7 02:00:33 LinuxServer systemd[1]: Finished Automatically configure NetworkManager in cloud.
Apr  7 02:06:33 LinuxServer systemd[1]: Starting Automatically configure NetworkManager in cloud..
.
Apr  7 02:06:33 LinuxServer systemd[1]: nm-cloud-setup.service: Deactivated successfully.
Apr  7 02:06:33 LinuxServer systemd[1]: Finished Automatically configure NetworkManager in cloud.
Apr  7 02:07:08 LinuxServer su[14658]: (to root) root on pts/0
Apr  7 02:09:13 LinuxServer su[14685]: (to splunk) root on pts/0
Apr  7 02:09:13 LinuxServer systemd[1]: Starting Hostname Service...
Apr  7 02:09:13 LinuxServer systemd[1]: Started Hostname Service.
Apr  7 02:09:43 LinuxServer systemd[1]: systemd-hostnamed.service: Deactivated successfully.
Apr  7 02:11:37 LinuxServer systemd[1]: Starting Automatically configure NetworkManager in cloud..
.
Apr  7 02:11:37 LinuxServer su[14721]: (to root) root on pts/0
Apr  7 02:11:37 LinuxServer systemd[1]: nm-cloud-setup.service: Deactivated successfully.
Apr  7 02:11:37 LinuxServer systemd[1]: Finished Automatically configure NetworkManager in cloud.
Apr  7 02:13:40 LinuxServer NetworkManager[698]: <info>  [1775528020.3897] dhcp4 (eth0): state cha
nged new lease, address=172.31.7.24
Apr  7 02:13:40 LinuxServer systemd[1]: Starting Network Manager Script Dispatcher Service...
Apr  7 02:13:40 LinuxServer systemd[1]: Started Network Manager Script Dispatcher Service.
Apr  7 02:13:40 LinuxServer systemd[1]: Starting Automatically configure NetworkManager in cloud..
.
Apr  7 02:13:40 LinuxServer systemd[1]: nm-cloud-setup.service: Deactivated successfully.
Apr  7 02:13:40 LinuxServer systemd[1]: Finished Automatically configure NetworkManager in cloud.
Apr  7 02:13:50 LinuxServer systemd[1]: NetworkManager-dispatcher.service: Deactivated successfull
y.
Apr  7 02:14:33 LinuxServer su[14778]: (to splunk) root on pts/0
Apr  7 02:14:33 LinuxServer systemd[1]: Starting Hostname Service...
Apr  7 02:14:33 LinuxServer systemd[1]: Started Hostname Service.
[splunk@LinuxServer ~]$
```

```
cat /var/log/audit/audit.log
```

if it still shows **"Permission denied" for audit.log**, then do the following:

```
chmod o+rx /var/log/audit
ls -ld /var/log/audit
setfacl -m u:splunk:r /var/log/audit/audit.log
getfacl /var/log/audit/audit.log
```

Expected:

```
user:splunk:r--
```

```
[root@LinuxServer ec2-user]# chmod o+rx /var/log/audit
[root@LinuxServer ec2-user]# ls -ld /var/log/audit
drwx---r-x. 2 root root 23 Mar 18 03:45 /var/log/audit
[root@LinuxServer ec2-user]# setfacl -m u:splunk:r /var/log/audit/audit.log
[root@LinuxServer ec2-user]# getfacl /var/log/audit/audit.log
getfacl: Removing leading '/' from absolute path names
# file: var/log/audit/audit.log
# owner: root
# group: root
user::rw-
user:splunk:r--
group::---
mask::r--
other::r--

[root@LinuxServer ec2-user]#
```

Verify that the audit.log file is readable:

```
sudo su - splunk
cat /var/log/audit/audit.log
```

```
type=USER_CMD msg=audit(1775529517.755:532): pid=14961 uid=0 auid=1000 ses=1 subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 msg='cwd="/home/ec2-user" cmd=7375202D2073706C756E6B exe="/usr/bin/sudo" terminal=pts/0 res=success'UID="root" AUID="ec2-user"
type=CRED_REFR msg=audit(1775529517.756:533): pid=14961 uid=0 auid=1000 ses=1 subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:setcred grantors=pam_env,pam_localuser,pam_unix acct="root" exe="/usr/bin/sudo" hostname=? addr=? terminal=/dev/pts/0 res=success'UID="root" AUID="ec2-user"
type=USER_START msg=audit(1775529517.758:534): pid=14961 uid=0 auid=1000 ses=1 subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:session_open grantors=pam_keyinit,pam_limits,pam_systemd,pam_unix acct="root" exe="/usr/bin/sudo" hostname=? addr=? terminal=/dev/pts/0 res=success'UID="root" AUID="ec2-user"
type=USER_AUTH msg=audit(1775529517.763:535): pid=14963 uid=0 auid=1000 ses=1 subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:authentication grantors=pam_rootok acct="splunk" exe="/usr/bin/su" hostname=? addr=? terminal=/dev/pts/0 res=success'UID="root" AUID="ec2-user"
type=USER_ACCT msg=audit(1775529517.764:536): pid=14963 uid=0 auid=1000 ses=1 subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:accounting grantors=pam_succeed_if acct="splunk" exe="/usr/bin/su" hostname=? addr=? terminal=/dev/pts/0 res=success'UID="root" AUID="ec2-user"
type=CRED_ACQ msg=audit(1775529517.764:537): pid=14963 uid=0 auid=1000 ses=1 subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:setcred grantors=pam_rootok acct="splunk" exe="/usr/bin/su" hostname=? addr=? terminal=/dev/pts/0 res=success'UID="root" AUID="ec2-user"
type=USER_START msg=audit(1775529517.767:538): pid=14963 uid=0 auid=1000 ses=1 subj=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 msg='op=PAM:session_open grantors=pam_keyinit,pam_keyinit,pam_limits,pam_systemd,pam_unix,pam_umask,pam_xauth acct="splunk" exe="/usr/bin/su" hostname=? addr=? terminal=/dev/pts/0 res=success'UID="root" AUID="ec2-user"
type=BPF msg=audit(1775529517.779:539): prog-id=110 op=LOAD
type=BPF msg=audit(1775529517.779:540): prog-id=111 op=LOAD
type=SERVICE_START msg=audit(1775529517.836:541): pid=1 uid=0 auid=4294967295 ses=4294967295 subj=system_u:system_r:init_t:s0 msg='unit=systemd-hostnamed comm="systemd" exe="/usr/lib/systemd/systemd" hostname=? addr=? terminal=? res=success'UID="root" AUID="unset"
[splunk@LinuxServer ~]$
```

6. To monitor the data, we need to create inputs.conf file in local folder of Search App in Splunk

```
cd /opt/splunkforwarder/etc/apps/search
```

1) Check if the local directory is present in Search app if not create a new directory named local

```
ls
mkdir local
cd local
```

```
[splunk@LinuxServer ~]$ cd /opt/splunkforwarder/etc/apps/search
[splunk@LinuxServer search]$ ls
default  metadata
[splunk@LinuxServer search]$ mkdir local
[splunk@LinuxServer search]$ cd local
[splunk@LinuxServer local]$ vi inputs.conf
```

2) Configure the inputs.conf file

```
vi inputs.conf
```

Press "**i**" or "**Insert**"

Enter the code below to monitor the file "/var/log/messages" & "/var/log/":

```
[monitor:///var/log/messages]
disabled = 0
sourcetype=linux_logs

[monitor:///var/log/audit/audit.log]
disabled = 0
sourcetype=linux_logs
```

```
~
~
```

Esc + ":wq!"

```
[monitor:///var/log/messages]
disabled = 0
sourcetype=linux_logs

[monitor:///var/log/audit/audit.log]
disabled = 0
sourcetype=linux_logs

~
~
~
~
~
~
~
~
:wq!
```

3) Restart Splunk

```
exit
/opt/splunkforwarder/bin/splunk restart
```

```
[splunk@LinuxServer local]$ exit
logout
[root@LinuxServer ec2-user]# /opt/splunkforwarder/bin/splunk restart
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunk:splunk /opt/splunkforwarder"
Stopping splunkd...
Shutting down.  Please wait, as this may take a few minutes.

Stopping splunk helpers...

Done.
splunkd.pid doesn't exist...

Splunk> The IT Search Engine.

Checking prerequisites...
        Checking mgmt port [8089]: open
        Checking conf files for problems...
        Done
        Checking default conf files for edits...
        Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-9.4
.9-03bb451d4e07-linux-amd64-manifest'
        All installed files intact.
        Done
All preliminary checks passed.

Starting splunk server daemon (splunkd)...
Done

[root@LinuxServer ec2-user]#
```

4) After making changes to inputs.conf file each time, use the below command to reload the configurations

```
/opt/splunkforwarder/bin/splunk _internal call
/services/data/inputs/monitor/_reload -auth
```

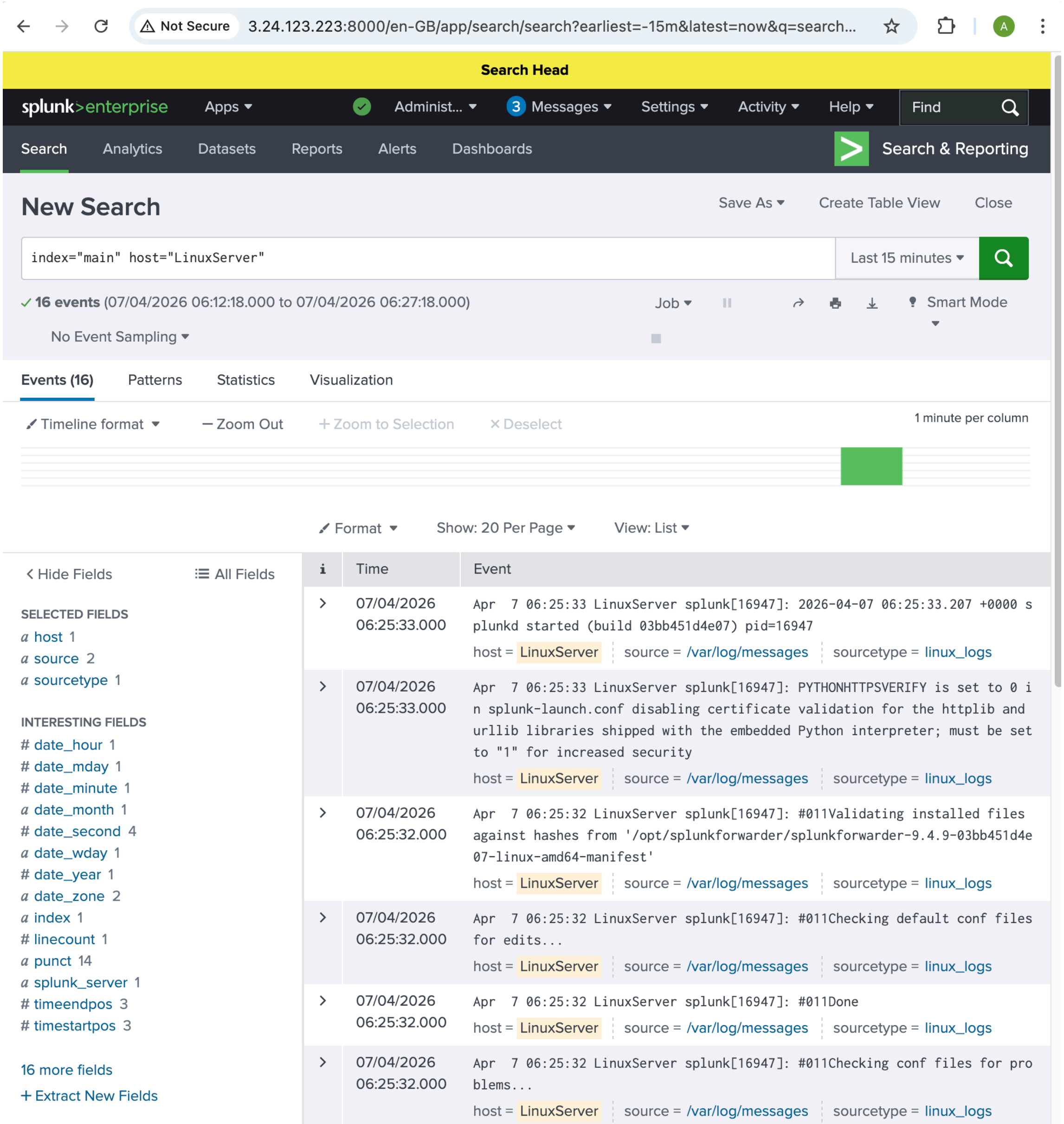
```
[splunk@LinuxServer ~]$ /opt/splunkforwarder/bin/splunk _internal call /services/data/inputs/mon
itor/_reload -auth
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R splunk:splunk /opt/splunkforwarder"
QUERYING: 'https://127.0.0.1:8089/services/data/inputs/monitor/_reload'
Splunk username: adminuf
Password:
HTTP Status: 200.
Content:
<?xml version="1.0" encoding="UTF-8"?>
<!--This is to override browser formatting; see server.conf[httpServer] to disable. . . . . . .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .
. . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . . .-->
<?xml-stylesheet type="text/xml" href="/static/atom.xsl"?>
<feed xmlns="http://www.w3.org/2005/Atom" xmlns:s="http://dev.splunk.com/ns/rest" xmlns:opensear
ch="http://a9.com/-/spec/opensearch/1.1/">
  <title>monitor</title>
  <id>https://127.0.0.1:8089/services/data/inputs/monitor</id>
  <updated>2026-04-07T04:48:08+00:00</updated>
  <generator build="03bb451d4e07" version="9.4.9"/>
  <author>
    <name>Splunk</name>
  </author>
  <link href="/services/data/inputs/monitor/_new" rel="create"/>
  <link href="/services/data/inputs/monitor/_reload" rel="_reload"/>
  <link href="/services/data/inputs/monitor/_acl" rel="_acl"/>
  <opensearch:totalResults>0</opensearch:totalResults>
  <opensearch:itemsPerPage>30</opensearch:itemsPerPage>
  <opensearch:startIndex>0</opensearch:startIndex>
  <s:messages/>
</feed>

[splunk@LinuxServer ~]$
```

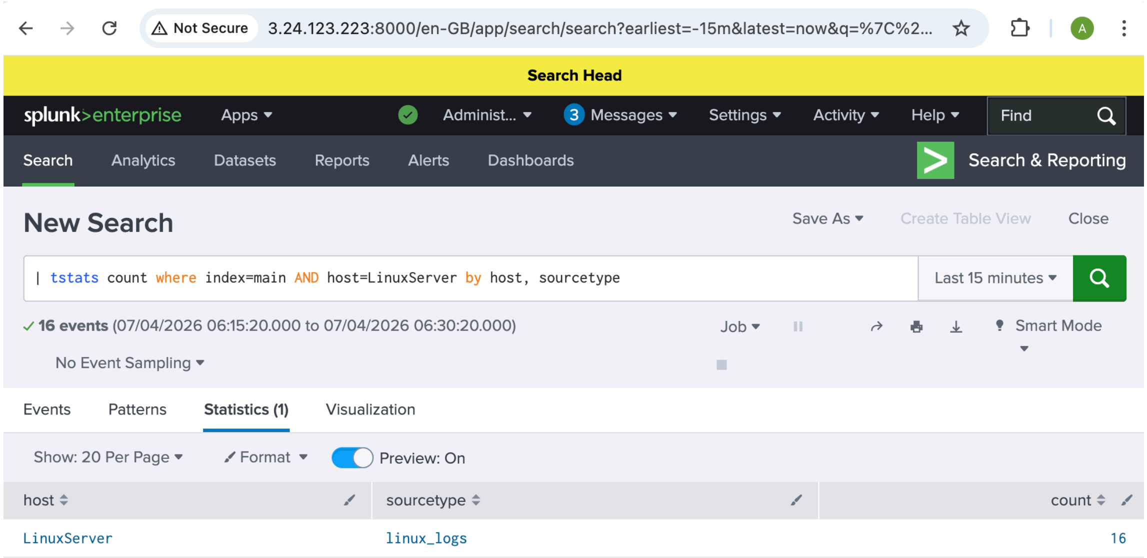
7. Verify the onboarded logs

In **SearchHead UI**, **Search and Reporting** app,

```
index=main host=<Server_Name>
```



```
| tstats count where index=main AND host=<Server_Name> by host,
sourcetype
```



8. Next Steps

Onboard Data from the Syslog-ng Server – Refer **08_Onboard_data_from_Syslog-ng_Server_V1.pdf.**